STAT 464 Assignment 2 Question 2

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2026-11-04

Set up time series

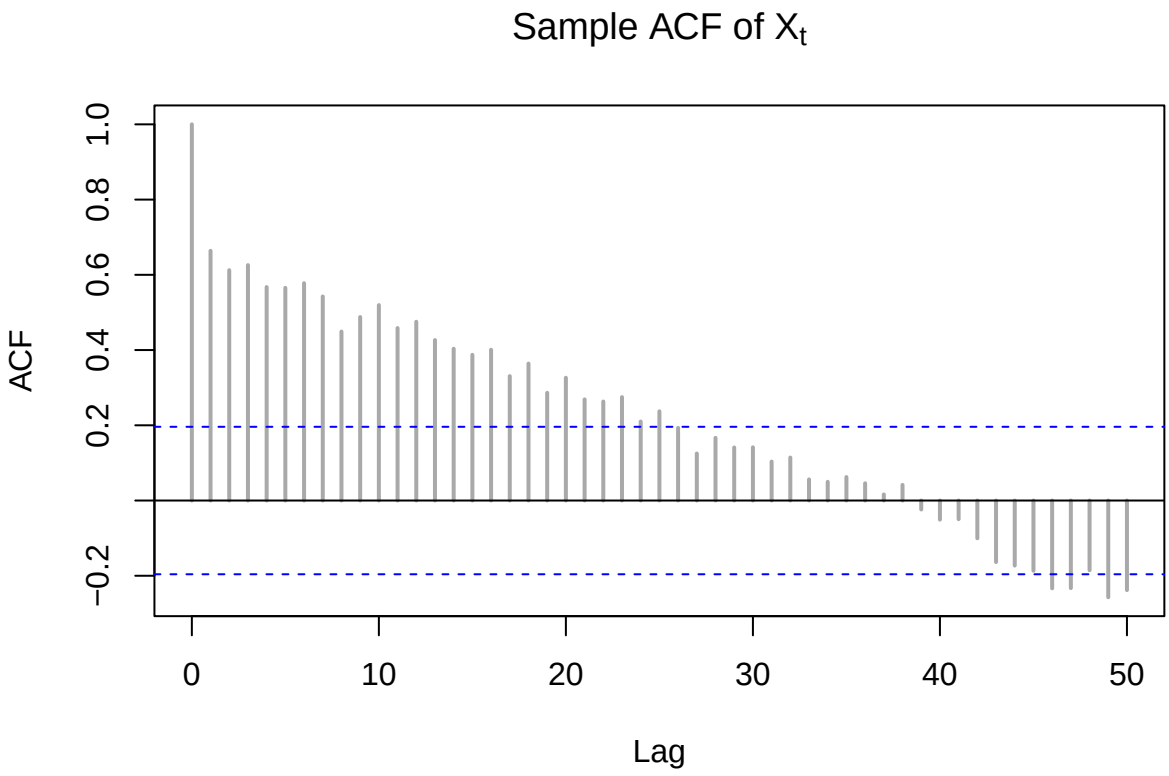
```
Z_t <- rnorm(100, 0, 2)
t <- c(1:100)

X_t <- 0.1*t + Z_t
```

ACF Plotting

```
h <- c(0:50)

acf(X_t, lag.max = 50,
    main = expression(paste("Sample ACF of ", X[t])),
    col = "darkgray", lwd = 2)
```



This ACF has the form it does by attempting to average towards 0 for the ACF when the lag `h != 0`. At `h = 0`, clearly we see that the ACF = 1 just like with the true/theoretical ACF computed. However, at `h != 0` it isn't exactly ACF = 0 like the theoretical formula, but trends towards it and likely will have some more negative lags `h > 50` to produce an average ~= 0.